System-Level Behavioral Modeling of a UCIe Transmitter

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Project Guide
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Overview about UCIe

- UCIe (Universal Chiplet Interconnect Express) is a high-speed interconnect standard for communication between chiplets inside a package.
- It enables multiple chiplets such as:
  - CPU
  - GPU
  - AI accelerators
  - Memory modules
    to communicate efficiently.
- UCIe provides:
  - High bandwidth
  - Low latency
  - Low power communication

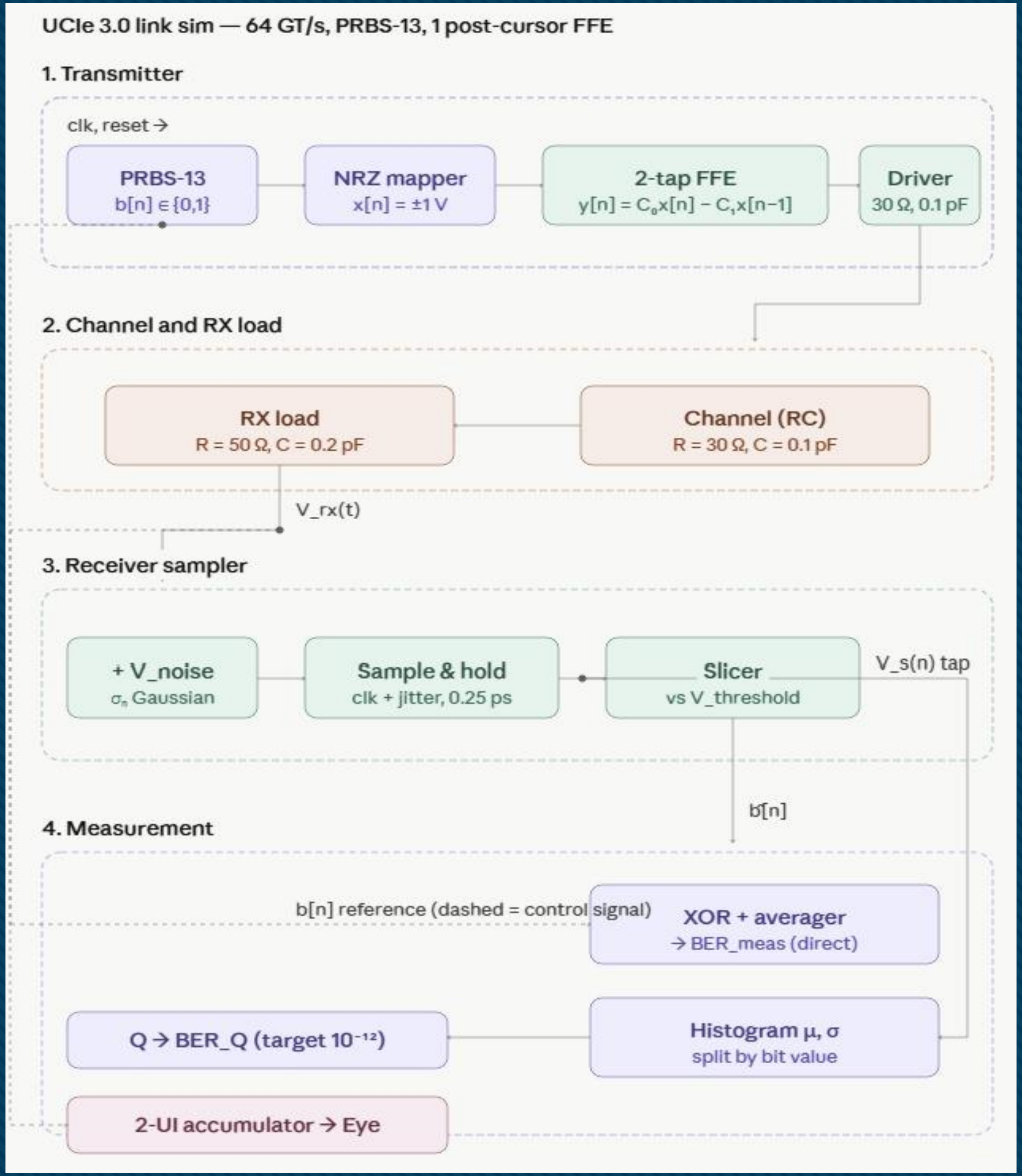
Problem Statement

- UCIe transfers data at very high speed.
- At high speed, the signal becomes weak and distorted in the channel.
- This can cause data errors during transmission.
- So, the transmitter must improve the signal before sending it.
- Problems such as:
    - Inter-Symbol Interference (ISI)
    - Channel loss
    - Noise

Project Objective

- To develop a MATLAB-based system-level model of a UCIe transmitter.
- To simulate high-speed data transmission in a UCIe link. To study the effects of:
  - channel loss
  - Inter-Symbol Interference (ISI)
  - Noise
- To implement Feed Forward Equalization for improving signal quality.
- To evaluate link performance using: eye diagram Bit Error Rate analysis

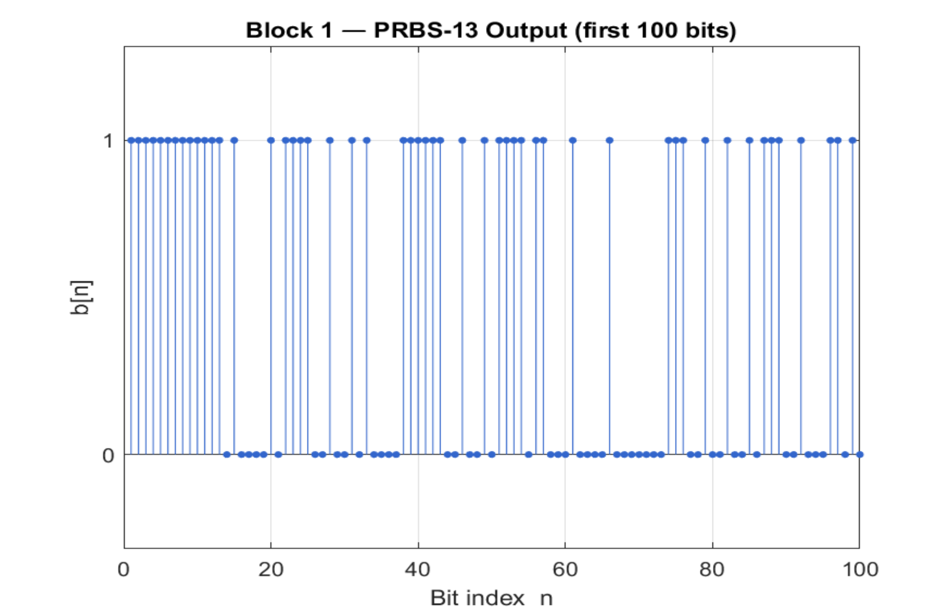
System Architecture



- Transmitter
  - Generates and processes the data signal.
- Channel and Rx load
  - Models signal loss during transmission.
- Receiver
  - Recovers the transmitted data.
- Measurement
  - Analyzes signal quality and transmission performance.

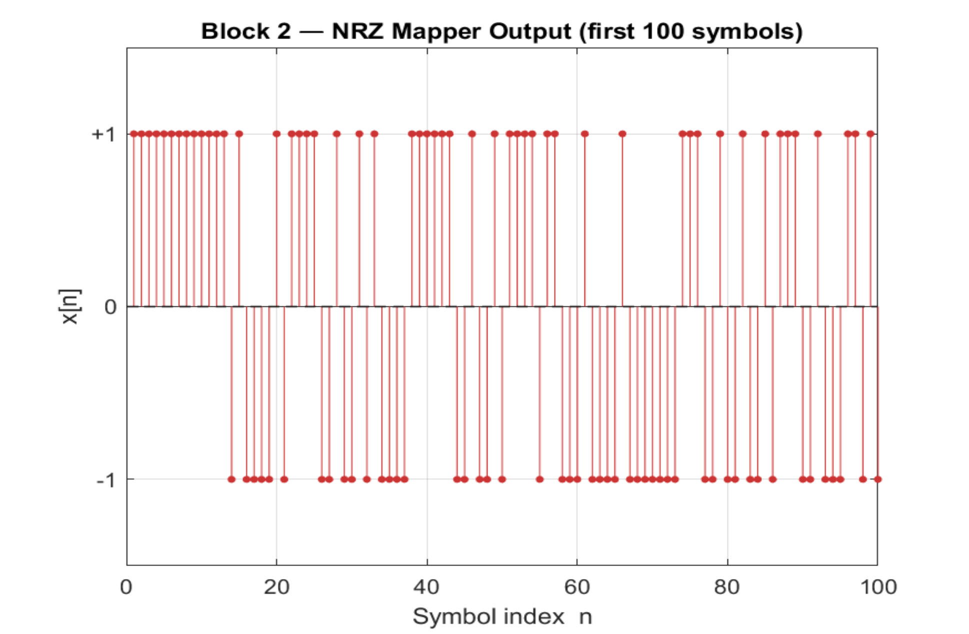
Pseudo Random Binary Sequence Generator

- PRBS is a pseudo-random binary sequence used as test data in communication systems.
- The PRBS generator provides realistic test data for evaluating the performance of the high-speed UCIe communication link.
- PRBS uses a Linear Feedback Shift Register (LFSR).
- LFSR is a group of shift registers where selected register bits are XOR to generate a new feedback bit. The bits shift continuously to create a pseudo-random sequence
- PRBS-13 uses fixed tap positions:
  - [13,4,3,1]
- The graph shows the first 100 bits generated by the PRBS-13 block.
- The output continuously changes between 0 and 1 in a random-looking pattern, which helps simulate realistic data transmission.



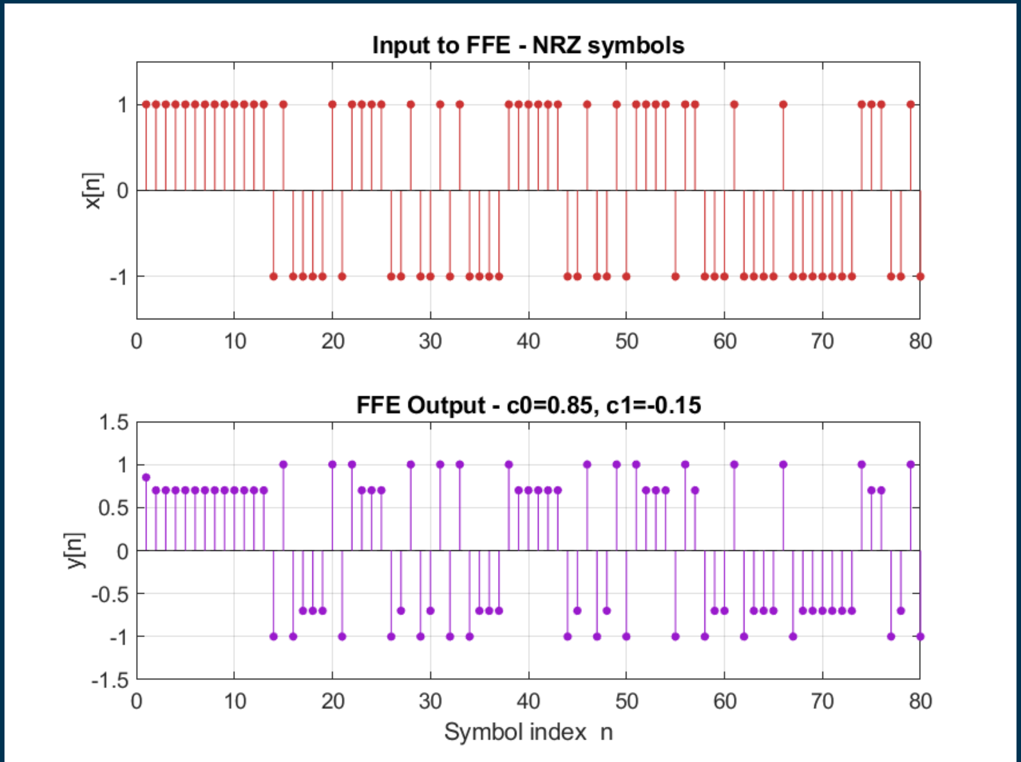
Non-Return-to-Zero(NRZ) Mapper

- NRZ (Non-Return-to-Zero) is a digital signaling method used in communication systems.
- It converts binary bits into voltage levels for transmission through the channel.
  - 1→+1
  - 0 →-1
- $x[n]=2b[n]-1$
- Communication channels transmit electrical signals, not digital bits directly.
- NRZ provides simple , low-complexity and efficient high-speed data transmission.
- NRZ uses bipolar signalling for better DC balance and improved signal integrity.



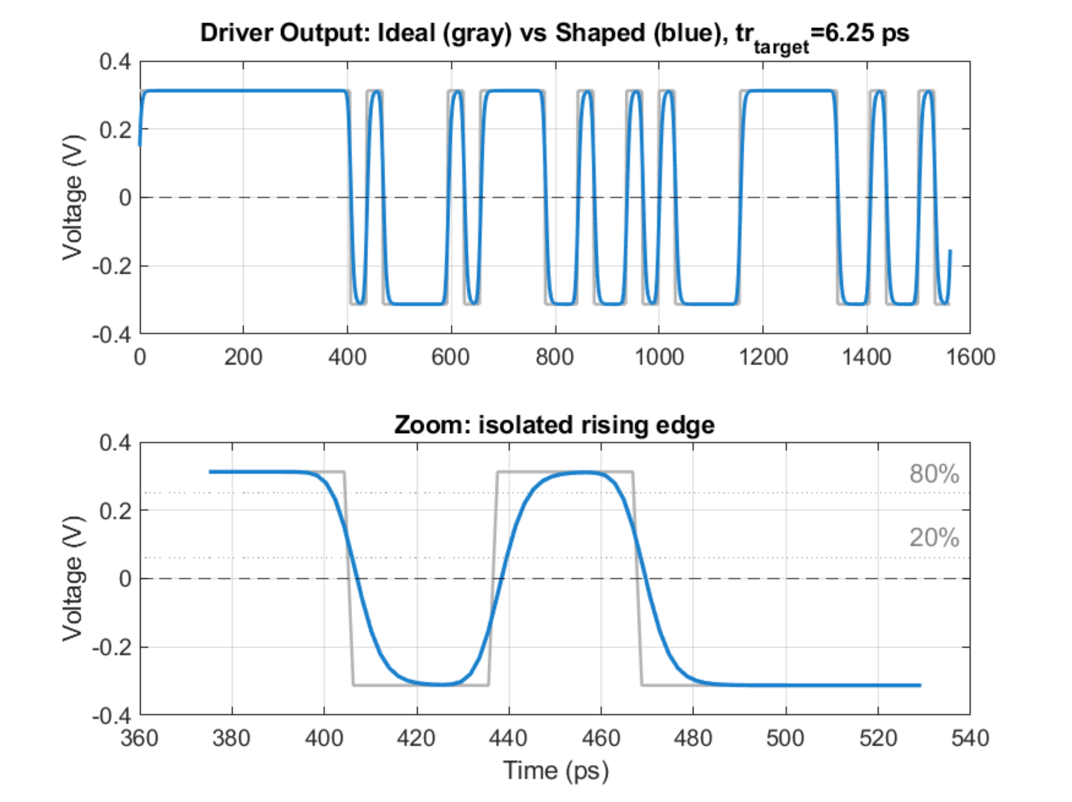
Feed Forward Equalizer (FFE)

- Feed Forward Equalizer (FFE) is a transmitter equalization technique used in high-speed communication systems.
- When data passes through the channel signal becomes distorted and previous bits affect current bits. This problem is called Inter-Symbol Interference (ISI)
- $y[n]=c0x[n]+c1x[n-1]$
- It modifies the transmitted signal using weighted tap coefficients.
- FFE improves signal quality and eye opening before transmission through the channel.



Driver Block

- The driver block converts the equalized digital symbols into a realistic analog transmission waveform before sending it through the channel.
- Upsampling
  - Converts symbol-rate data into waveform samples.
  - Creates a continuous-time waveform for transmission.
- Voltage Scaling : Vswing=0.625V
  - +1→+0.3125V
  - −1→−0.3125V
- Rise/Fall Shaping
  - Real hardware cannot change voltage instantly.
- RC Output Modeling
  - RC output modeling simulates the resistance and capacitance present in the real transmitter output stage.



The Channel Block (Multi-Die Interconnect)

- The Channel represents the physical microscopic copper trace routing signals from the Transmitter die to the Receiver die.
- When ultra-high-speed data (32–64 GT/s) pass and Skin Effect. This causes Inter-Symbol Interference (ISI).
- It acts as a continuous-time Low-Pass Filter, mathematically modeled as a discrete exponential moving average based on a
    - Time constant: $\tau = R_{ch} \times C_{ch}$ (using standard $R_{ch} = 30\ \Omega$, $C_{ch} = 0.1\ pF$)
- It modifies the transmitted signal by erasing high-frequency transitions (like alternating 0101 bits), resulting in a calculated -3dB bandwidth cutoff (e.g., around 16 to 28 GHz depending on the test case).
- The Channel is the primary source of signal degradation, causing severe vertical eye closure and horizontal jitter, which the FFE is explicitly designed to pre-correct.

RX Load Block (Receiver Front-End)

- The RX Load represents the electrical environment at the receiver pad boundary.
- Impedance mismatch at the receiver causes:
- Signal Reflection
- Backward energy propagation
- Data corruption
- Models a parallel termination network:
- Zload =50Ω||0.2 pF
- Signal modification mechanisms:
- 50 Ω resistor absorbs incoming energy
- 0.2 pF pad capacitance steals incoming charge
- Parasitic capacitance slows signal transitions:
- Rise edge slump
- Fall edge slump
- Final unavoidable analog degradation before receiver sampling

Noise ,Jitter and Slicer

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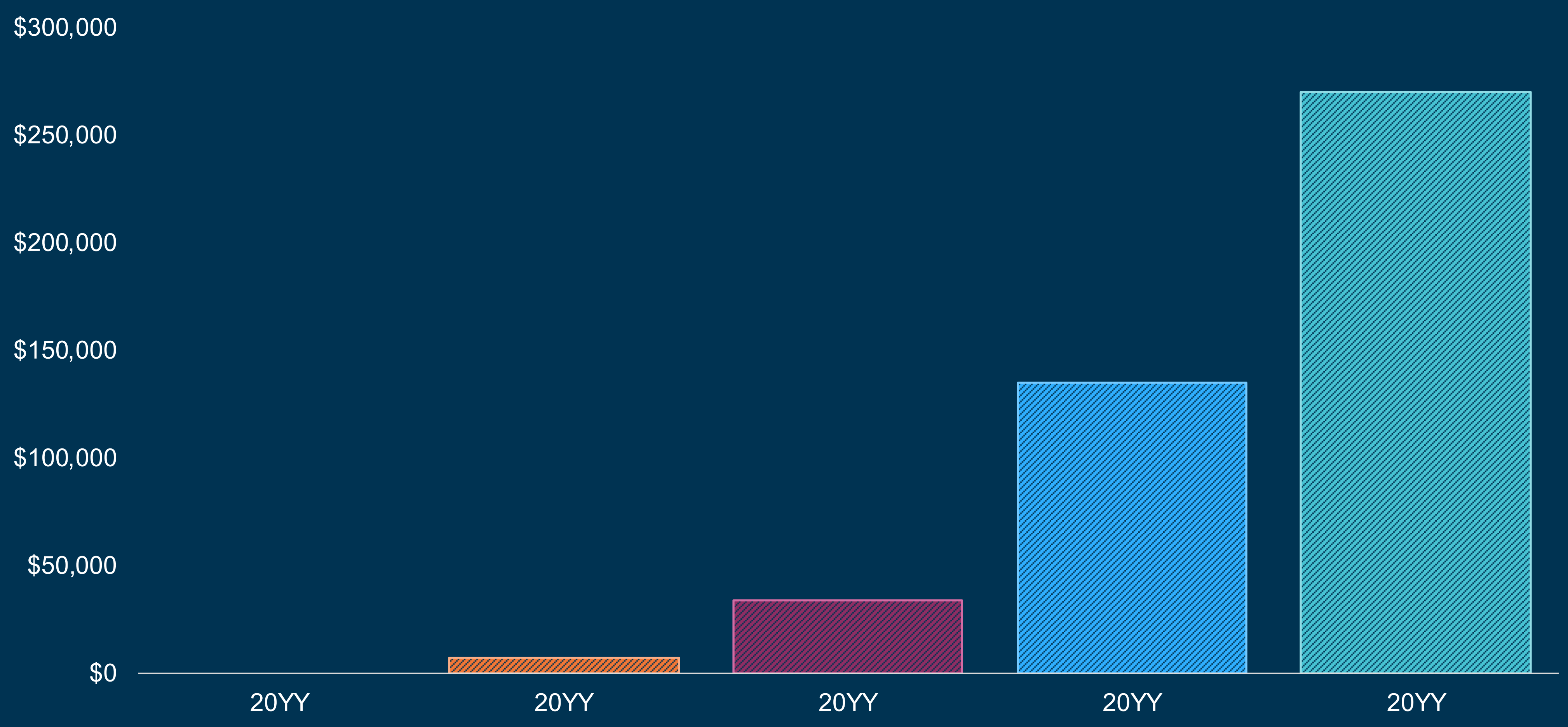

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